

MAT102 Midterm Cheat Sheet

SYMBOL KEY (keep this open while you study)

Symbol	Meaning
$\in / \notin$	is / is not an element of
$\subseteq / \subsetneq$	subset / proper subset
$\emptyset$	empty set
$\cup / \cap / \setminus / ^c / \Delta / \times$	union / intersection / difference / complement / symmetric difference / product
$P(A)$	power set of A
$\wedge / \vee / \neg / \Rightarrow / \Leftrightarrow$	and / or / not / implies / iff
$\forall / \exists$	for all / there exists
$\mathbb{N} / \mathbb{Z} / \mathbb{Q} / \mathbb{R}$	naturals / integers / rationals / reals
$\leq, \sim$	generic custom-defined relation — always re-check the definition given, never assume standard meaning
$a b$	a divides b
$[x]$	equivalence class of x
$A/\sim$	quotient set of A by $\sim$
$f^{-1}(B)$	preimage of B (a set), NOT “1 over f”

1. PRODUCT SETS

Definition: $A \times B = \{(x,y) : x \in A, y \in B\}$. Order matters — $(a,b) \neq (b,a)$ in general.

Tricks: - $|A \times B| = |A| \cdot |B|$. - $A \times B = B \times A$ **only if** $A=B$, or one of them is empty. - To prove two product sets are equal, or to prove a subset relationship: **pick an arbitrary element (x,y), show it satisfies membership on both sides.** - Distributes over $\cup$ and $\cap$: $A \times (B \cup C) = (A \times B) \cup (A \times C)$, same for $\cap$. These are usually TRUE — test them by picking an arbitrary (x,y) and unpacking the logic ($\vee$ and $\wedge$ distribute over each other, which is why these work). - $(A \setminus B) \times C = (A \times C) \setminus (B \times C)$ — TRUE, again check via arbitrary element + logic.

Proof template ($A \times B = B \times A \Rightarrow A=B$): 1. Let $a \in A$ be arbitrary. Since $B \neq \emptyset$, pick any $b \in B$. 2. $(a,b) \in A \times B = B \times A$, so $(a,b) \in B \times A$, meaning $a \in B$. 3. This shows $A \subseteq B$. Symmetric argument (start with $b \in B$) shows $B \subseteq A$. 4. Conclude $A=B$.

2. POWER SETS

Definition: $P(A)$ = set of ALL subsets of A (including $\emptyset$ and A itself).

Tricks: - $|A|=n \Rightarrow |P(A)|=2^n$. Always the fastest way to double check you enumerated everything. - $P(\emptyset) = \{\emptyset\}$ — this has **1 element**, not 0. Classic trap. - Nesting ladder: $P(\emptyset)=\{\emptyset\}$ (1 elt) $\rightarrow P(P(\emptyset))=\{\emptyset, \{\emptyset\}\}$ (2 elts) $\rightarrow P(P(P(\emptyset)))$ has 4 elements. Each step doubles. - **$\in$ vs $\subseteq$ trap:** if $x \in A$, then $x \in P(A)$ is FALSE and $\{x\} \in P(A)$ is TRUE. Always ask “is this a subset of A?” not “is this in A?” - $P(A \cap B) = P(A) \cap P(B)$ is **TRUE** (a subset of $A \cap B$ is automatically a subset of both). - $P(A \cup B) = P(A) \cup P(B)$ is **FALSE** in general (a subset can mix elements from both A and B without being wholly inside either) — find a counterexample with small sets to confirm on the test. - $P(A) \subseteq P(B) \Leftrightarrow A \subseteq B$ is **TRUE** both directions — useful biconditional.

3. RELATIONS

Definition: Any $R \subseteq A \times B$ is a relation. Homogeneous = $A=B$ (this is 95% of your test).

Mechanical check for “is $(a,b) \in R$?”: plug a,b into the defining condition, evaluate true/false. No shortcuts — just do the arithmetic/logic exactly as written.

Common relation patterns you’ll see: - $n \mid (a-b)$: “same remainder mod n ” — usually gives an equivalence relation. - $a-b \geq 0$: gives an order (not symmetric). - Divisibility $a \mid b$: reflexive, NOT symmetric ($2 \mid 4$ but $4 \nmid 2$), transitive $\rightarrow$ gives a partial order, not equivalence.

4. REFLEXIVE / SYMMETRIC / TRANSITIVE

The exact check — do this every time, don’t skip steps:

- **Reflexive:** list every element $x \in A$. Confirm $(x,x) \in R$ for ALL of them. One missing = fails.
- **Symmetric:** for every (a,b) actually in R , confirm (b,a) is also in R . One missing reverse = fails.
- **Transitive:** find every chain $(a,b), (b,c) \in R$ (second coord of one matches first coord of next). Confirm $(a,c) \in R$ for all chains. One missing shortcut = fails.

Speed tip: you only need ONE counterexample to disprove a property. Don’t check everything if you find a break early — write it down and move to the next property.

Common trap: relations defined with “+” instead of “-” (e.g. $4 \mid (a+b)$ instead of $4 \mid (a-b)$) usually break reflexivity. Read definitions letter by letter.

5. EQUIVALENCE RELATIONS

Definition: reflexive AND symmetric AND transitive, all three, no exceptions.

Proof template for “show R is an equivalence relation”: 1. **Reflexive:** plug (a,a) into the definition, show it’s always true (usually reduces to something trivially true like $x=x$). 2. **Symmetric:** assume the defining equation/condition holds for (a,b) . Algebraically rearrange to show the condition holds for (b,a) too. 3. **Transitive:** assume the condition holds for (a,b) and (b,c) . Add/combine the two equations algebraically, cancel the shared variable, land on the condition for (a,c) .

Worked skeleton (equation-based relations): - Given: $a+d=b+c$ (this is the $(a,b) \sim (c,d)$ condition from your sheet) - Reflexive: $a+b=b+a$ ✓ (commutativity) - Symmetric: $a+d=b+c \iff c+b=d+a$ (just swap sides) ✓ - Transitive: add $(a+d=b+c)$ and $(c+f=d+e)$, cancel c,d from both sides $\rightarrow a+f=b+e$ ✓

6. EQUIVALENCE CLASSES

Definition: $[x] = \{y \in A : x \sim y\}$.

Two free facts to always cite: 1. $[x]$ is never empty ($x \in [x]$ by reflexivity). 2. $[x]=[y] \iff x \sim y$. If $x \not\sim y$, then $[x] \cap [y] = \emptyset$ (completely disjoint).

How to describe an equivalence class on the test: substitute the defining condition with one variable fixed. E.g. for $x \sim y := [x]=[y]$, $[x] = \{y : [y]=[x]\} =$ the interval $[x], [x]+1)$.

7. QUOTIENT SET / QUOTIENT SPACE

Definition: $A/\sim = \{[x] : x \in A\}$ — the set of all equivalence classes.

Key facts: - $A/\sim \subseteq P(A)$ (each class is a subset of A). - The classes always **partition** A : disjoint, and their union is all of A . - Diagonal relation ($x \sim y := x=y$) $\implies$ every class is a singleton $\{x\} \implies A/\sim$ is basically

a copy of A . - Universal relation ($x \sim y$ always true) $\implies$ one giant class = A itself $\implies A/\sim = \{A\}$, just one element.

On the test, “describe the quotient space” means: identify what the equivalence classes look like in general (a formula or shape), then describe the set of all of them (often “in bijection with” some simpler set like $\mathbb{Z}$ or an interval).

8. INDUCED FUNCTION ON QUOTIENT SPACE

The setup: $f: A \rightarrow B$ and equivalence relation $\sim$ on A . Define $\bar{f}([x]) := f(x)$.

The one thing that can go wrong — well-definedness: since $[x]$ can be written using any representative, you must check: $x \sim x' \implies f(x) = f(x')$. If this fails, $\bar{f}$ isn't a valid function.

Proof template: 1. Suppose $x \sim x'$. 2. Use the definition of $\sim$ to derive some relationship between x and x' . 3. Show this relationship forces $f(x) = f(x')$. 4. Conclude $\bar{f}$ is well-defined.

9. INJECTIVITY / SURJECTIVITY / BIJECTIVITY

Definitions (memorize word-for-word): - **Injective (one-to-one):** $\forall x, y \in A, f(x) = f(y) \implies x = y$. (Different inputs give different outputs — no two inputs share an output.) - **Surjective (onto):** $\forall b \in B, \exists a \in A$ such that $f(a) = b$. (Every element of B is hit by something.) - **Bijective:** both injective and surjective (perfect pairing — inverse function exists).

How to prove injective — direct method: Assume $f(x) = f(y)$. Do algebra to force $x = y$. *Example:* $f(x) = 2x + 3$. Assume $2x + 3 = 2y + 3 \implies 2x = 2y \implies x = y$. Injective. ✓

How to prove injective — contrapositive method (useful when direct algebra is messy): Assume $x \neq y$. Show $f(x) \neq f(y)$.

How to DISPROVE injective: find ONE pair $x \neq y$ with $f(x) = f(y)$. *Example:* $f(x) = x^2$ on $\mathbb{R}$. $f(2) = 4 = f(-2)$, but $2 \neq -2$. Not injective.

How to prove surjective: Take arbitrary $b \in B$. **Construct** an explicit $a \in A$ (usually by solving $f(a) = b$ for a) and verify $f(a) = b$. *Example:* $f(x) = 2x + 3$ on $\mathbb{R} \rightarrow \mathbb{R}$. Given $b \in \mathbb{R}$, let $a = (b - 3)/2$. Check: $f(a) = 2 \cdot (b - 3)/2 + 3 = b$. ✓ Surjective.

How to DISPROVE surjective: find one $b \in B$ with no preimage. *Example:* $f(x) = x^2$ on $\mathbb{R} \rightarrow \mathbb{R}$. $b = -1$ has no real square root. Not surjective (codomain matters here!).

Critical trap: injectivity/surjectivity depend on the stated domain AND codomain, not just the formula. $f(x) = x^2$ is not surjective $\mathbb{R} \rightarrow \mathbb{R}$, but IS surjective $\mathbb{R} \rightarrow [0, \infty)$. Always check what the codomain is stated to be.

Counting functions: number of functions from A to B ($|A| = m$, $|B| = n$) is n^m . If asked for injective functions specifically: $n \cdot (n - 1) \cdot (n - 2) \cdots (n - m + 1)$ (only valid if $n \geq m$).

HOW TO WRITE A SOLID PROOF (general strategy)

1. **State what you're assuming and what you need to show**, explicitly, before doing algebra. E.g. “Assume $x \sim y$. We want to show $f(x) = f(y)$.”
2. **“Show $A \subseteq B$ ” almost always means:** let $a \in A$ be arbitrary, show $a \in B$. Don't skip the word “arbitrary” — it's what makes the proof general.
3. **“Show $A = B$ ” means prove both $A \subseteq B$ and $B \subseteq A$ separately**, unless you have a chain of $\iff$ (iff) statements throughout.
4. **For “true or false, justify” questions:** if TRUE, give a general proof (arbitrary element argument). If FALSE, give ONE explicit counterexample with actual numbers/sets — don't just gesture at why it might fail.

5. **Never skip the reflexive/symmetric/transitive labels** when checking equivalence relations — write which property you're checking before you check it, so partial credit is possible even if you get stuck.
 6. **When defining an equivalence class or quotient set, use set-builder notation exactly:** $[x] = \{y \in A : \text{condition}\}$, don't just describe it in words alone.
 7. **Watch quantifier order** — " $\forall y \exists x$ " is very different from " $\exists x \forall y$." Read your problem sheet's Q4 definitions (S vs T) as a live example of this exact trap.
 8. **If stuck on a proof for >5 minutes on the actual test, write down the definition and what you'd need to show, then move on** — partial credit for correct setup beats a blank page.
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10. LEFT/RIGHT INVERSES (simplified)

Plain idea: a left-inverse "undoes" f when you start from A . A right-inverse "undoes" f when you start from B .

- **g is a left-inverse of f** means: $g(f(x)) = x$ for every $x \in A$. Written $g \circ f = \text{id}_A$.
- **h is a right-inverse of f** means: $f(h(y)) = y$ for every $y \in B$. Written $f \circ h = \text{id}_B$.

The one-line connection to memorize: f is injective $\iff f$ has a left-inverse. f is surjective $\iff f$ has a right-inverse.

Why this makes intuitive sense: if f is injective, no two inputs share an output, so you can always trace an output back to its unique input — that trace-back IS the left-inverse. If f is surjective, every output has at least one input hitting it — just pick one such input for each output, and that choice-function IS the right-inverse.

How to actually construct one on the test: solve $y=f(x)$ for x in terms of y — that formula, written as a function of y , is your inverse (left or right depending on direction you're going).

Example: $f: [0, \infty) \rightarrow \mathbb{R}$, $f(x)=x^2$. Left-inverse: $g(y)=\sqrt{y}$ (only defined sensibly using nonnegative root since domain is $[0, \infty)$). Check: $g(f(x))=\sqrt{x^2}=x$ ✓ for $x \geq 0$.

11. PREIMAGE/IMAGE $\leftrightarrow$ INJECTIVE/SURJECTIVE (simplified)

Two clean facts, plug-and-check style:

- f is **injective** $\iff$ for every subset $C \subseteq A$, $f^{-1}(f(C)) = C$. (Plain idea: take a chunk of A , push it forward then pull it back — if f is injective, you land exactly where you started, nothing extra sneaks in.)
- f is **surjective** $\iff$ for every subset $D \subseteq B$, $f(f^{-1}(D)) = D$. (Plain idea: take a chunk of B , pull it back then push it forward — if f is surjective, you recover the whole chunk, nothing is missing.)

Memory trick: injective protects you going forward-then-back (nothing extra added). Surjective protects you going backward-then-forward (nothing missing).

12. COMPOSITION PROPERTIES (simplified)

Let $f: A \rightarrow B$ and $g: B \rightarrow C$, $h = g \circ f$.

Two clean, provable-by-direct-proof facts: - If f and g are **both injective**, then $h = g \circ f$ is **injective**. - If f and g are **both surjective**, then $h = g \circ f$ is **surjective**. - (Together: both bijective $\implies$ composition bijective.)

Quick proof shape for injective $\circ$ injective: Assume $h(x_1) = h(x_2)$, i.e., $g(f(x_1)) = g(f(x_2))$. Since g is injective, this forces $f(x_1) = f(x_2)$. Since f is injective, this forces $x_1 = x_2$. Done.

Quick proof shape for surjective $\circ$ surjective: Let $z \in C$ be arbitrary. Since g is surjective, $\exists y \in B$ with $g(y) = z$. Since f is surjective, $\exists x \in A$ with $f(x) = y$. Then $h(x) = g(f(x)) = g(y) = z$. Done.

Trap to watch for: the REVERSE direction isn't automatic — "if h is injective, are f and g each injective?" only half holds (f must be injective, but g doesn't have to be — g just needs to behave injectively on the specific outputs f actually produces). Always test reverse-direction claims with a small counterexample before assuming they're true.

LAST HONEST NOTE

This sheet is a reference, not a study session. It will not move a single point onto your transcript by existing — only by you re-deriving these proofs cold, without looking, will they actually transfer. Use this to check your work, not to replace doing the work.